



Type 2 Diabetes in Children and Adolescents

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Type 2 diabetes mellitus is a disorder of insulin resistance, leading to an insulin release that is not enough for the body's metabolic needs. Type 2 diabetes is characterized by high blood sugar and is often associated with overweight or obesity.

- Resistance to the effects of insulin often occurs before symptoms, and is often related to obesity or the metabolic syndrome.
- Early symptoms are related to hyperglycemia (high blood glucose level) and include excessive thirst, excessive hunger, excessive urination, and blurred vision.
- Doctors diagnose diabetes by measuring blood sugar levels.
- Diabetes can damage blood vessels and increase the risk of heart attack, stroke, chronic kidney disease, and vision loss.
- Diabetes can damage nerves and cause problems with the sense of touch.
- People with type 2 diabetes need to follow a healthy diet that is low in refined carbohydrates (including sugar), saturated fat, and processed foods. They also need to exercise and maintain a healthy weight. Many also need to take medication to lower their blood glucose levels.

Among children, type 2 diabetes occurs mainly in adolescents but is becoming increasingly common among younger children with overweight (weigh more than 85% of children of similar age, sex, and height) or obesity (weigh more than 95% of children of similar age, sex, and height). About one-third of children newly diagnosed with diabetes have type 2 diabetes, and about three-quarters of children with type 2 diabetes have obesity.

Type 2 diabetes typically develops after puberty has begun.

Children at higher risk of developing type 2 diabetes include those who:

- Have overweight or obesity
- Have Native American, Canadian First Nation, Indigenous Australian, Black, Hispanic, East and south Asian, Middle Eastern, and Pacific Islander heritage

- Have [high blood pressure](#), [high blood levels of lipids](#) (fats), [obstructive sleep apnea](#), dark and thick skinfolds on the nape of the neck (acanthosis nigricans), fatty liver, [polycystic ovary syndrome](#) (PCOS), or a [small-for-gestational-age birth weight](#)
- Are male
- Have close relatives (parent or sibling) with type 2 diabetes
- Have a mother who had diabetes while pregnant ([gestational diabetes](#)) or who has a history of type 2 diabetes
- Excessive intake of sugar-sweetened beverages
- Are not physically active

Symptoms of Type 2 Diabetes in Children and

Many children do not have any symptoms or have only mild symptoms, and their type 2 diabetes may be detected only when blood or urine tests are done for other reasons (such as during a physical before playing sports or going to camp).

Symptoms in children with type 2 diabetes are milder than those in type 1 diabetes and develop more slowly. Parents may notice an increase in the child's thirst and urination or only vague symptoms, such as fatigue.

Some people with type 2 diabetes develop skin darkening and thickening around the back of the neck and in the armpits, called acanthosis nigricans.

Acanthosis Nigricans

IMAGE



Acanthosis nigricans is darkening and thickening of skin caused by insulin resistance. It is often seen on the back of the neck and in the armpits of children and adolescents with type 2 diabetes.

PHOTOS PROVIDED BY THOMAS HABIF, MD.

Children with type 2 diabetes are less likely to develop [ketoacidosis](#) than those with type 1 diabetes, but ketoacidosis or a type of severe dehydration and confusion (called hyperosmolar hyperglycemic state) can still develop.

Diagnosis

The diagnosis of type 2 diabetes is made based on an elevated blood glucose level (see [Diagnosis of Diabetes in Children and Adolescents](#)), and after evaluating for the type of diabetes (type 1, type 2, or another type [see [Determining the Type of Diabetes](#)]). Patients will also undergo a blood test for diabetes-specific antibodies, which are usually not found in type 2 diabetes.

Testing after diagnosis

Children who are diagnosed with type 2 diabetes have blood tests to determine how their liver and kidneys are functioning and urine tests. At diagnosis, children who have type 2 diabetes are also tested for other problems, such as [high blood pressure](#), [high blood levels of lipids](#) (fats), and [steatotic liver disease](#) (previously called fatty liver), because these problems are common among children with type 2 diabetes. Other tests are done depending on symptoms. For example, children with daytime sleepiness or who snore while sleeping are tested for [obstructive sleep apnea](#), and adolescent girls who are hairy and have acne or menstrual irregularities are tested for [polycystic ovary syndrome](#).

Treatment

Type 2 diabetes treatment

In type 2 diabetes, lifestyle modifications focus on weight in the majority of children. Steps to improve food choices and manage food intake include eliminating sugary drinks, controlling portion size, switching to low-fat foods, and increasing fiber by eating more fruits and vegetables.

Choosing healthy foods can help control blood glucose and protect heart health. Children should focus on eating fruits and vegetables, whole grains, and high-fiber foods (for example, foods that have at least 3 grams of fiber or more per serving). Food should generally not contain many highly processed (refined) [carbohydrates](#), particularly candy, baked goods (such as cookies, donuts, and pastries), and sugary drinks. Children should have no more than 4 to 8 ounces of 100% fruit juice per day. They should avoid regular soda, sweetened iced tea, lemonade, fruit punch, and sports drinks altogether. Children also should avoid [foods high in saturated fats](#), such as baked goods, snack foods (such as potato chips and corn tortilla chips), deep-fried foods (such as french fries), and fast food. Some of these foods may still contain [trans fats](#), common ingredients in certain commercial foods, which are being removed because they have been shown to be associated with an increased risk of heart disease.

Regular exercise is important because it improves glucose control and makes it easier to lose weight. Because vigorous exercise can cause a significant drop in blood glucose, some children with type 1 diabetes may need to consume some extra carbohydrates before and/or during a workout.

Children with type 2 diabetes usually are not treated in the hospital unless the diabetes is severe. Usually they are given medications to lower blood glucose levels (antihyperglycemic medications) at a regular doctor's office visit. Children with severe diabetes may need to be hospitalized to start insulin treatment. Less commonly, children with type 2 diabetes develop severe dehydration or, as in type 1 diabetes, [diabetic ketoacidosis](#) (DKA).

Metformin is most commonly the first medication given by mouth (orally) for children and adolescents under 18 years of age. It is started at a low dose and often increased over several weeks to higher doses. It should be taken with food or given as an extended-release formulation to prevent nausea and abdominal pain.

Insulin is given to children who are hospitalized because of ketosis, DKA, or hyperosmolar hyperglycemic state. Insulin can often be stopped after several weeks once glucose levels return to normal after treatment with metformin. Children whose type 2 diabetes is not controlled by metformin alone are given insulin or another medication called liraglutide. About half of adolescents with type 2 diabetes ultimately require insulin.

Liraglutide, **exenatide**, and **dulaglutide** are injectable medications that can be given to children over 10 years of age who have type 2 diabetes. **Semaglutide** is another injectable medication that can be given to people 12 years of age and over to manage type 2 diabetes and to treat obesity. These medications are known as GLP-1 agonists. GLP-1 is a hormone that has several roles in the body, including preventing more glucose from going into the bloodstream, slowing stomach emptying, and

affecting the areas of the brain that process hunger and fullness (satiety). The GLP-1 agonists work by acting like the GLP-1 hormone and therefore help manage blood glucose by triggering the pancreas to release more insulin and by having an effect that reduces appetite and hunger, resulting in weight loss. GLP-1 agonists also help lower HbA1C levels. They may be given to children who are taking metformin but whose HbA1C level is not in the target range, or they can be given instead of metformin to children who cannot tolerate that medication.

Empagliflozin is a medication taken by mouth that can be given to children over 10 years of age with type 2 diabetes. This medication is a sodium-glucose cotransporter-2 (SGLT2) inhibitor. Empagliflozin helps lower blood glucose levels by increasing the amount of glucose that is removed from the body in urine. It cannot be taken by people who have severe kidney disease or who are on [dialysis](#). It can increase the risk of DKA and may cause urinary tract infections (UTIs) and genital yeast infections.

Other medications used for adults with type 2 diabetes may help some adolescents, but they are more expensive, and there is limited evidence for their use in children.

Some children who lose weight, improve their food choices, and exercise regularly may be able to stop taking the medications.

Screening and Prevention

Type 2 diabetes

Because prompt treatment (such as changes in food choices, an increase in physical activity, and weight loss) may help prevent or delay the start of type 2 diabetes, [children at risk of type 2 diabetes](#) should be screened with a blood test that measures the hemoglobin A1C levels. This test should first be done when children are 10 years old or when puberty starts (if puberty occurred at a younger age) and should be repeated every 3 years if normal.

Some risk factors for type 2 diabetes can be modified. For example, children who are obese should lose weight, and all children should get regular exercise (see [Nutrition and exercise](#)).